INCH-POUND

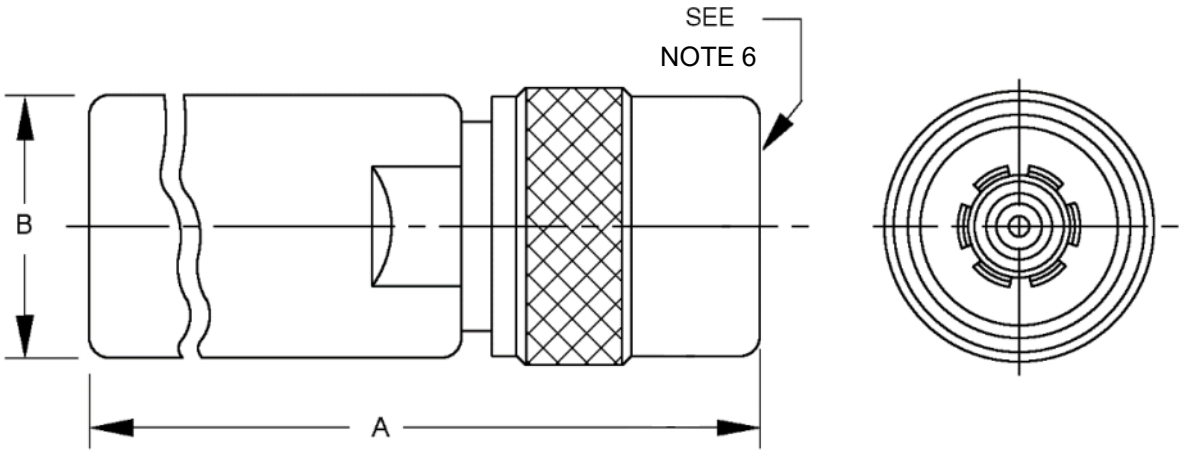
MIL-PRF-39012/6E
25 April 2024
SUPERSEDING
MIL-PRF-39012/6D
w/AMENDMENT 5
28 May 2021

PERFORMANCE SPECIFICATION SHEET

CONNECTORS, PLUG, ELECTRICAL, COAXIAL, RADIO FREQUENCY, (SERIES C (CABLED), MALE, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-39012.



NOTES:
1 For dimensions A and B, see tables I and III.
2 Dimension B is the largest overall diameter of the connector.
3 Wrench flats to accommodate standard wrench in accordance with FED-STD-H28.
4 Dimension A defines the maximum length of the connector when assembled to the appropriate cable.
5 All undimensioned pictorial configurations are for reference purposes only.
6 Series C, pin contact interface, in accordance with MIL-STD-348.

FIGURE 1. General Configuration.

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TABLE I. Dash numbers, cross-reference, and dimensions.

Dash Number 1/	Applicable cable M17/ 2/	Typical mating connector M39012/ (optional hardware) 3/	Dimensions	Inches 13/ (millimeters) maximum
CATEGORY A – FIELD SERVICEABLE (NO SPECIAL TOOLS REQUIRED)				
0001 4/	73-RG212 5/ 112-RG304	7-0002 8-0002, 11-0001	A B	1.609 (40.87) .781 (19.84)
0002 4/	65-RG165 74-RG213 75-RG214 5/ 86-00001 6/	7-0001 8-0001 11-0002	A B	1.609 (40.87) .781 (19.84)
0003 4/	78-RG217 5/		A B	1.875 (47.63) .891 (22.63)
0004 4/	72-RG211 5/		A B	2.250 (57.15) 1.328 (33.73)
0005 4/	79-RG218 5/		A B	2.250 (57.15) 1.328 (33.73)
0013 4/	74-RG215 5/	7-0010 8-0009, 11-0010	A B	2.250 (57.15) .891 (22.63)
0014 8/	73-RG212 5/ 112-RG304	7-0018 8-0016, 11-0018	A B	1.609 (40.87) .781 (19.84)
0015 8/	65-RG165 74-RG213 75-RG214 5/ 86-00001 6/	7-0017 8-0015 11-0019		
0016 8/	78-RG217 5/		A B	1.875 (47.63) .891 (22.63)
0017 8/	72-RG211 5/		A B	2.250 (57.15) 1.328 (33.73)
0018 8/	79-RG218 5/			
0019 8/	74-RG215 7/	7-0019 8-0017, 11-0020	A B	2.250 (57.15) .891 (22.63)
0020 8/	92-RG115 6/	7-0020 8-0018, 11-0021	A B	1.609 (40.87) .781 (19.84)
CATEGORY C – FIELD REPLACEABLE (SAE AS22520 CRIMP TOOL) SEE NOTE NEXT TO APPLICABLE CABLE FOR CRIMP DIE 8/ 9/ 14/				
0027	73-RG212 5/ 10/ 112-RG304 10/	7-0014 8-0013, 11-0012	A	2.000 (50.80) .781 (19.84)
0028	74-RG213 5/ 11/ 65-RG165 6/ 11/	7-0011 8-0010, 11-0013		
0029	75-RG214 5/ 11/	7-0012 8-0011, 11-0014		
0030	86-00001 5/ 6/ 11/	7-0013 8-0012, 11-0015		
0031	6-RG11 5/ 11/ 12/	7-0015, 11-0016		
0032	92-RG115 6/ 11/	7-0016 8-0014, 11-0017		

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1/ For cross reference of dash number to superseded Part or Identifying Number (PIN) or type designation, see table IV.
2/ The latest version of each cable shall be applicable.
3/ Optional hardware M39012/25-0003.
4/ Inactive for new design.
5/ Cable to be used when performing tests requiring cable except as in 6/ and 12/.
6/ Cable to be used for the +200°C temperature cycling tests.
7/ Armored cable.
8/ These connectors have captivated center contacts.
9/ Category C connectors are assembled using the applicable crimp tool in accordance with SAE AS22520, to the specified cable stripped as shown on figure 2.
10/ M22520/5-35 closure A or M22520/5-55 closure A.
11/ M22520/5-61.
12/ These are not 50 ohm cables; therefore, when attached to the specified connectors, VSWR, RF leakage and insertion loss are not applicable.
13/ Complete connector assembly shall consist of a body, center contact, ferrule, and assembly instructions.

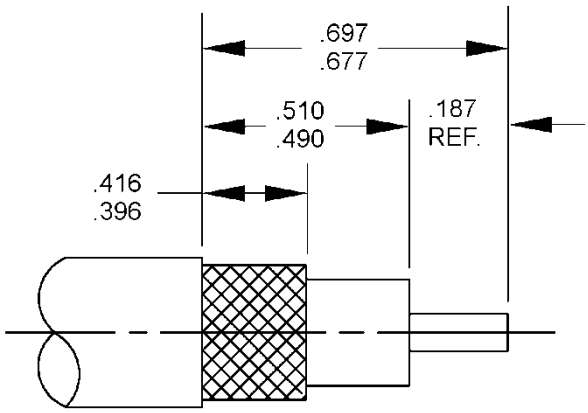


FIGURE 2. Recommended cable stripping dimensions for category C connectors.

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ENGINEERING INFORMATION:

Nominal impedance: 50 ohms.
Frequency range: 0 to 11 GHz.
Voltage rating: 1,000 volts rms maximum working voltage at sea level.
250 volts rms maximum at 70,000 feet (4.437 kPa).
Temperature rating: -65°C to +165°C.

REQUIREMENTS:

Dimensions and configuration: See figure 1 and MIL-STD-348.
Force to engage and disengage:
Longitudinal force: 4 1/2 pounds, maximum.
Torque: 4 inch-pounds, maximum (.45 Nm).

Coupling proof torque: Not applicable.
Inspection conditions: Coupling torque not applicable.

Mating characteristics: See MIL-STD-348.

Outer contact:

Test ring ID: .411 inch (10.44 mm) maximum, 16 microinches (0.406 µm) finish.
Insertion force: 15 pounds (66.72 N), maximum.
Insertion depth: .125 inch (3.18 mm), minimum.

Hermetic seal: Not applicable.
Leakage (pressurized connectors): Not applicable.
Insulation resistance: In accordance with MIL-STD-202-302, condition B, 5,000 mega ohms, minimum.
Center contact retention: 6 pounds (26.69 N), minimum axial force.
Corrosion (salt spray): In accordance with MIL-STD-202-101, condition B.
Voltage standing wave ratio (VSWR): From 500 MHz to 11 GHz, or approximately 80 percent of the upper cutoff frequency of the cable, whichever is lower; 1.35, maximum.

Swept frequency VSWR setup:

Item 6: VSWR shall be less than 1.015 +.005F (F in GHz).
Item 16: VSWR shall be less than 1.015 +.005F (F in GHz).
Second step of VSWR checkout procedure: VSWR shall be less than 1.045 +.015F (F in GHz).
Group B inspection: VSWR shall be less than 1.10 +.01F (F in GHz).
Qualification and group C inspection: VSWR shall not exceed 1.15.

Connector durability: 500 cycles, minimum at 12 cycles per minute, maximum. The connector shall meet mating characteristics and force to engage and disengage requirements.

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Contact resistance: In milliohms, maximum:

	Initial	After environment
Center contact:	1.0	1.5
Outer contact:	.15	Not applicable
Braid to body:	.05	Not applicable

Dielectric withstanding voltage at sea level: In accordance with MIL-STD-202-301, 3,000 volts rms minimum at sea level.

Vibration, high frequency: In accordance with MIL-STD-202-204, test condition B.

Shock: In accordance with MIL-STD-202-213, test condition I.

Thermal shock: In accordance with MIL-STD-202-107, test condition B, except test high temperature shall be +85°C. Test high temperature shall be +200°C for connectors using +200°C cables (see tables I and III).

Moisture resistance: In accordance with MIL-STD-202-106. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Corona level:

Voltage: 750 volts rms, minimum.
Altitude: 70,000 feet.

RF high potential withstanding voltage:

Voltage and frequency: 2,500 volts rms tested at a frequency from 5 to 7.5 MHz.
Leakage current: Not applicable.

Cable retention force:

Non-crimp assemblies: 75 pounds (333.62 N), minimum.
Crimp assemblies:

50 lbs. (222.41 N) minimum for cables .155 -.189 inch (3.94 – 4.80 mm) OD.
60 lbs. (266.90 N) minimum for cables .190 - .229 inch (4.83 – 5.82 mm) OD.
75 lbs. (333.62 N) minimum for cables .230 - .249 inch (5.84 – 6.32 mm) OD.
90 lbs. (400.34 N) minimum for cables .250 inch (6.35 mm) OD and larger.

Coupling mechanism retention force: 100 pounds (444.82 N), minimum.

RF leakage: -55 dB minimum, tested at frequency between 2 and 3 GHz.

RF insertion loss:

.15 dB maximum at 9 GHz.

$.05\sqrt{F\,(GHz)}$ dB maximum tested at 3 and 6 GHz.

Part or Identifying Number (PIN): M39012/06- (dash number from table I or B number from table III).

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QUALIFICATION: Qualification is not required. Conformance inspection shall consist of a visual and mechanical inspection in accordance with MIL-PRF-39012 and the requirements stated in this specification sheet. Conformance inspection shall consist of the inspections specified in table II.

Samples for conformance inspection shall be representative of the products proposed to be furnished to this specification. Test samples shall consist of 100% testing, up to 12 connectors. Test samples shall be examined and tested in accordance with MIL-PRF-39012 and this specification. Test results shall be submitted to the acquiring activity. Periodic inspection is not required.

TABLE II. Conformance inspection.

Examination or test	Requirement paragraph	Test method paragraph
Group I		
Visual and mechanical examination		
Material	3.3	4.6.1.1
Finish	3.3.1	
Dissimilar metals	3.3.2	
Configuration and features (dimensions)	3.4	4.6.1.2
Marking	3.29	
Mating (visual indication)	3.4.1	4.6.1.1
Force to engage/disengage		
Bayonet and threaded types	3.5.1	4.6.2.1
“Push-on” connector types	3.5.2	4.6.2.2
Mating characteristics	3.7	4.6.4
Permeability of nonmagnetic materials	3.8	4.6.5
Workmanship	3.30	4.6.1
Insulation resistance	3.11	4.6.8
Group II		
Center contact retention	3.12	4.6.9
Corrosion	3.13	4.6.10
Group III		
VSWR (cabled)	3.14	4.6.11
Connector durability	3.15	4.6.12
Safety wire hole pullout	3.32	4.6.25
Group IV		
Contact resistance	3.16	4.6.13
Dielectric withstanding voltage	3.17	4.6.14
Vibration	3.18	4.6.15
Shock (specified pulse)	3.19	4.6.16
Thermal shock	3.20	4.6.17
Moisture resistance	3.21	4.6.18
Corona level	3.22	4.6.19
RF high potential withstanding voltage	3.23	4.6.20
Cable retention force	3.24	4.6.21
Coupling mechanism retention force	3.25	4.6.22
Group V		
RF leakage	3.26	4.6.23
Group VI		
RF insertion loss	3.27	4.6.24
Group VII		
Contact resistance	3.16	4.6.13

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TABLE III. Category B – non-field replaceable (special tools may be required).

Not for Air Force, Army, or Navy use. For OEM use only.

PIN 2/ 3/ M39012/06	Applicable Cable M17/ 4/	Typical mating connector M39012/ (optional hardware) 1/	Dimensions	Inches 8/ (millimeters) maximum
B0021	73-RG212 5/ 112-RG304	7-0024 8-0022	A B	2.000 (50.80) .781 (19.84)
B0022	74-RG213 5/ 65-RG165 6/	7-0021 8-0019 11-0023		
B0023	75-RG214 5/	7-0022 8-0020 11-0024		
B0024	86-00001 5/ 6/	7-0023 8-0021 11-0025		
B0025	6-RG11 5/ 7/	7-0025 11-0026		
B0026	92-RG115 6/	7-0026 8-0023 11-0027		

1/ Optional hardware M39012/25-0001.
2/ For cross-reference of PIN to superseded PIN or type designation, see table IV.
3/ These connectors have captivated center contacts.
4/ The latest version of each cable shall, be applicable.
5/ Cable to be used when performing tests requiring cable except as in 6/ and 7/.
6/ Cable to be used for the +200°C temperature cycling tests.
7/ These are not 50 ohm cables; therefore, when attached to the specified connectors, VSWR, RF leakage, and insertion loss are not applicable.

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Supersession data: See table II.

TABLE IV. Supersession data. 1/ 2/

Preferred PIN M39012/6	Superseded PINs or superseded type designation
-0001	UG-626/U
-0002	UG-573/U
-0003	UG-707/U
-0004	UG-711/U
-0005	UG-708/U
-0013	UG-943/U
-0014	M39012/06-0001
-0015	M39012/06-0002
-0016	M39012/06-0003
-0017	M39012/06-0004
-0018	M39012/06-0005
-0019	M39012/06-0013
-0020	
B0021	M39012/06-0021, M39012/06-0007, UG-1748/U
B0022	M39012/06-0022, M39012/06-0008, UG-1749/U
B0023	M39012/06-0023, M39012/06-0009, UG-1750/U
B0024	M39012/06-0024, M39012/06-0010, UG-1751/U
B0025	M39012/06-0025, M39012/06-0011, UG-1752/U
B0026	M39012/06-0026
-0027	M39012/06-0021
-0028	M23329/1-01, M23329/1-03
-0029	M23329/1-02
-0030	M23329/1-04
-0031	M23329/1-09
-0032	

1/ The superseded PIN or the type designation is for reference only. Where a superseded PIN or type designation is not given; none was assigned or will be assigned. The PIN M39012/06-XXXX shall be used in all cases for marking and identifying the connector.

2/ The basic type designation includes all letter versions of the specified number, e.g. UG-18/U includes UG-18 A/U, UG-18B/U, etc.

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TABLE V. Maintenance replacements for category B.

Category B number 1/	Category C dash number	Category A dash number
B0021	0027	0001
B0022	0028	0002
B0023	0029	-------
B0024	0030	-------
B0025	0031	-------
B0026	0032	0020

1/ Category B connectors are for original installation only. They will not be stocked or procured by the government.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Referenced documents. In addition to MIL-PRF-39012, this document references the following:

FED-STD-H28
MIL-STD-202-101
MIL-STD-202-106
MIL-STD-202-107
MIL-STD-202-204
MIL-STD-202-213
MIL-STD-202-301
MIL-STD-202-302
MIL-STD-348
SAE AS22520

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2024-133)

Review activities:
Army – AT, CR4
Navy – AS, MC, OS
Air Force – 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil .

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